DSML SECOND SERIES

Complete Exam Notes & Question Practice

20MCA201 — Data Science & Machine Learning

Module 3: Regression methods onwards | Module 4: Neural network learning | Module 5: Up to ROC curve

Prepared for exam revision

Includes exam-ready theory, key formulas, solved teacher-paper numericals, PYQ-style practice, and a last-minute revision checklist.

How to Use These Notes

These notes combine the teacher-provided October 2025 second-series paper, the 2021 paper pasted in chat, the uploaded DSML PYQs, and the portions explicitly confirmed by your teacher. Past-paper recurrence helps prioritise practice but cannot guarantee the same questions will appear.

Scope: Module 3 from regression methods onwards; Module 4 neural network learning; Module 5 through ROC curves. SVM appears in the teacher's paper and older PYQs, but your stated Module 4 portion only names neural network learning. Confirm SVM coverage; it is included here as a clearly marked optional topic.

Exam writing pattern for a 15-mark answer

• 1. Definition and objective

• 2. Main concept, terminology and diagram where useful

• 3. Equations/formulas with each symbol defined

• 4. Algorithm or step-by-step method

• 5. Worked example or numerical

• 6. Advantages, limitations and conclusion

Master Formula Sheet

Topic	Formula
Simple regression	■ = a + bX
OLS objective	SSE = Σ(Y■ – ■■)² = Σ(Y■ – a – bX■)²
Slope	b = Σ[(X■–X■)(Y■–■)] / Σ[(X■–X■)²]
Intercept	a = ■ – bX■
Pearson correlation	r = Σ[(X■–X■)(Y■–■)] / √(Σ(X■–X■)² Σ(Y■–■)²)
Multiple regression	■ = a + b■X■ + … + b■X■
Neuron	z = Σw■x■ + b; output y = ϕ(z)
Weight update	w■ ← w■ – η ∂L/∂w■
Bias update	b ← b – η ∂L/∂b
Euclidean distance	d(x,c) = √Σ■(x■–c■)²
Precision	TP / (TP + FP)
Recall / TPR / Sensitivity	TP / (TP + FN)
Specificity	TN / (TN + FP)
F1-score	2 × Precision × Recall / (Precision + Recall)
FPR	FP / (FP + TN) = 1 – Specificity

MODULE 3 — Regression Methods

1. Regression: definition, objective and applications

Regression is a supervised learning/statistical method that models the relationship between a dependent variable and one or more independent variables. It is commonly used to predict a continuous numerical outcome. Examples include predicting examination marks from study hours, house prices from property features, or sales from advertising expenditure.

Simple linear regression uses one independent variable. **Multiple linear regression** uses two or more independent variables. The dependent variable is the quantity to be predicted; independent variables are the inputs used for prediction.

2. Simple linear regression

```
■ = a + bX
```

Here ■ is the predicted dependent value, a is the intercept, b is the slope, and X is the independent variable. The slope represents the expected change in predicted Y for a one-unit increase in X. The intercept is the predicted Y at X = 0.

3. Ordinary Least Squares (OLS)

OLS estimates regression coefficients by choosing the line that minimises the sum of squared residuals. A residual is the difference between an observed value and its predicted value.

```
e■ = Y■ − ■■
SSE = Σ(Y■ − ■■)² = Σ(Y■ − a − bX■)²
```

The squared residuals prevent positive and negative errors from cancelling and penalise large errors more heavily. The values of a and b are chosen to make SSE as small as possible.

Coefficient formulas

```
b = Σ[(X■ − X■)(Y■ − ■)] / Σ[(X■ − X■)²]
a = ■ − bX■
```

Equivalent computational slope formula:

```
b = [nΣXY − (ΣX)(ΣY)] / [nΣX² − (ΣX)²]
```

OLS procedure: (1) arrange X and Y observations, (2) calculate means or required sums, (3) calculate slope, (4) calculate intercept, (5) write the regression equation, and (6) substitute the given X to predict Y. If a question explicitly says Y is independent and asks for the regression equation, reverse the direction and predict X from Y.

4. Teacher-paper numerical 6: predict X from Y

Given X = 55, 60, 65, 70, 80 and Y = 52, 54, 56, 58, 62. The question states Y is the independent variable, so use X■ = a + bY.

Y	X	Y²	YX
52	55	2704	2860
54	60	2916	3240
56	65	3136	3640
58	70	3364	4060
62	80	3844	4960
ΣY=282	ΣX=330	ΣY²=15964	ΣYX=18760

```
b = [5(18760) − 330(282)] / [5(15964) − 282²] = 740/296 = 2.5
X■ = 330/5 = 66; ■ = 282/5 = 56.4
a = X■ − b■ = 66 − 2.5(56.4) = −75
```

Answer: X■ = −75 + 2.5Y. The data fit this equation exactly.

5. Teacher-paper numerical 7: predict statistics grade

X = 95, 85, 80, 70, 60; Y = 85, 95, 70, 65, 70. Predict Y when X = 80. Here X is independent and Y is dependent.

```
n=5; ΣX=390; ΣY=385; ΣX²=31350; ΣXY=30475
b = [5(30475) − 390(385)] / [5(31350) − 390²]
b = 2300/6000 = 0.38333
X■=78; ■=77; a = 77 − 0.38333(78) = 47.10
■ = 47.10 + 0.38333X; at X=80, ■ ≈ 77.77
```

Predicted statistics grade ≈ 77.77. Keep adequate decimal precision until the final step.

6. Correlation and Pearson’s coefficient

Correlation measures the direction and strength of association between two variables. Pearson’s correlation coefficient r lies between –1 and +1.

```
r = Σ[(X■–X■)(Y■–■)] / √{Σ(X■–X■)² Σ(Y■–■)²}
```

r = +1 means perfect positive linear correlation; r = –1 means perfect negative linear correlation; r = 0 means no linear correlation. Correlation alone does not establish causation.

Correlation versus regression

Correlation	Regression
Measures strength and direction of association.	Models a relationship and can predict values.
Treats variables symmetrically.	Distinguishes dependent and independent variables.
r is between –1 and +1.	Regression coefficients are not restricted to this interval.
Does not establish causation.	Prediction also does not establish causation.

7. Multiple linear regression

Multiple linear regression predicts one dependent variable using two or more independent variables.

```
■ = a + b■X■ + b■X■ + … + b■X■
```

For example, a student's mark can be predicted using study hours, attendance and internal assessment marks. Each coefficient represents the expected change in predicted Y for a one-unit change in that variable, holding other included variables constant.

```
β■ = (X■X)■¹X■Y (when X■X is invertible)
```

Advantages of OLS: simple, interpretable, minimises squared residuals, and supports prediction. Limitations: sensitive to outliers; a linear model can underfit nonlinear data; correlated predictors can destabilise coefficients; inference depends on assumptions.

MODULE 4 — Neural Network Learning

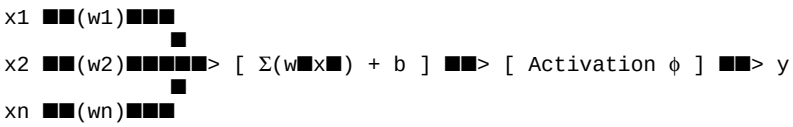
1. Artificial neuron

An artificial neuron is a computational unit inspired by a biological neuron. It receives inputs, multiplies them by weights, adds a bias, and applies an activation function to produce an output.

```
z = Σ(w■x■) + b
y = ϕ(z) = ϕ(Σw■x■ + b)
```

For zero bias, b = 0, so y = ϕ(Σw■x■). Here x■ are inputs, w■ are weights, b is bias, ϕ is the activation function, and y is the output.

Diagram to reproduce in exam



Weights determine the influence of inputs. Bias shifts the activation threshold. The activation function introduces nonlinearity, allowing networks to model complex relationships.

2. Activation functions

An activation function transforms the neuron's weighted sum into its output.

```
Step: ϕ(z)=1 if z≥0, otherwise 0
Sigmoid: ϕ(z)=1/(1+e^(−z))
Tanh: ϕ(z)=(e^z−e^(−z))/(e^z+e^(−z))
ReLU: ϕ(z)=max(0,z)
```

Step is used in basic threshold units; sigmoid produces values between 0 and 1; tanh produces values between –1 and +1; ReLU is commonly used in hidden layers because it is simple and helps with gradient flow for positive inputs. Softmax is often used to convert output scores into class probabilities for multiclass classification.

```
Softmax: p■ = e^(z■) / Σ■ e^(z■)
```

3. Network topology

Network topology refers to the arrangement of neurons and connections. A feedforward network typically contains an input layer, one or more hidden layers, and an output layer.

• Input layer: receives features. • Hidden layers: transform inputs using weights, biases and activation functions. • Output layer: produces the prediction or class scores.

Networks may be shallow or deep, fully connected or sparsely connected. Training adjusts weights and biases to reduce a loss function.

4. Backpropagation learning algorithm

Backpropagation trains a neural network by calculating how the loss changes with each weight and bias, then propagating error gradients from the output layer back through earlier layers using the chain rule.

Algorithm

• Initialise weights and biases, usually with small random values.

• Forward pass: compute each layer's weighted sums and activations.

• Calculate loss by comparing predicted output with the target.

• Backward pass: calculate gradients of the loss using the chain rule.

• Update weights and biases using a learning rate.

• Repeat over training examples/epochs until a stopping criterion is reached.

Weight and bias updates

```
w■ ← w■ − η(∂L/∂w■)
b ← b − η(∂L/∂b)
```

Here η is the learning rate and L is the loss. The gradients indicate how changing each parameter affects the loss. The negative gradient direction is used to reduce the loss locally.

For a single neuron with activation y = ϕ(z), z = Σw■x■+b, and loss L, define δ = ∂L/∂z. Then:

```
∂L/∂w■ = δx■; ∂L/∂b = δ
```

For a sigmoid neuron with squared-error loss L = ½(y–t)², δ = (y–t)y(1–y). For other activations or losses, the derivative changes. In a multilayer network, the chain rule propagates the gradients backward layer by layer.

Strengths: learns complex patterns; supports many prediction tasks; parameters can be trained from examples. Limitations: may require substantial data and computation; performance depends on architecture and hyperparameters; can overfit; training can be sensitive to initialisation and learning rate.

OPTIONAL TOPIC — SVM AND KERNEL TRICK

Your teacher's 2025 paper and older PYQs include SVM, but your confirmed Module 4 wording says neural network learning. Verify with your teacher whether SVM is included before prioritising this section.

A Support Vector Machine (SVM) is a supervised learning algorithm that finds a decision boundary (hyperplane) separating classes, ideally with the largest possible margin. Support vectors are training points that lie closest to the boundary and influence its position.

A linear hyperplane can be written as:

```
w■x + b = 0
```

Linearly separable data can be separated by a straight line in 2D (or hyperplane in higher dimensions). Concentric circles are not linearly separable by a straight line.

Kernel trick

The kernel trick allows an SVM to represent similarity in a higher-dimensional feature space without explicitly calculating the transformed coordinates. This can produce nonlinear decision boundaries in the original input space.

Examples of kernels:

```
Linear: K(x,z)=x■z

Polynomial: K(x,z)=(γx■z+r)^d

RBF: K(x,z)=exp(−γ||x−z||²), γ>0
```

For concentric-circle data, an RBF kernel is a suitable example because it can model a nonlinear boundary. The choice of kernel and its parameters should be validated for the dataset.

MODULE 5 — Clustering and Model Evaluation

1. K-means clustering

K-means is an unsupervised learning algorithm that partitions data into K clusters. Each point is assigned to the nearest centroid, and each centroid is updated to the mean of the points assigned to its cluster.

Algorithm

• Choose K initial centroids.

• Calculate each point's distance to every centroid.

• Assign each point to the nearest centroid.

• Recalculate each centroid as the mean of its assigned points.

• Repeat assignment and update until assignments/centroids stabilise or the stopping condition is met.

Euclidean distance

```
d(x,c) = √[(x■–c■)² + (x■–c■)² + … + (x■–c■)²]
```

For 2D points, d((x■,y■),(x■,y■)) = √[(x■–x■)²+(y■–y■)²]. The centroid of a cluster is the coordinate-wise mean of its points.

Choosing K: the Elbow Method

Within-Cluster Sum of Squares (WCSS) measures the sum of squared distances between each point and its assigned centroid:

```
WCSS = Σ■ Σ_{x■∈C■} ||x■–μ■||²
```

As K increases, WCSS cannot increase when clusters are optimally assigned, and it typically decreases. The Elbow Method plots K against WCSS and selects a K near the point where the rate of decrease becomes much smaller—the elbow. The elbow is a heuristic, not a guarantee of a uniquely correct K.

2. Older PYQ: K-means one epoch

Points: A1=(2,10), A2=(2,5), A3=(8,4), A4=(5,8), A5=(7,5), A6=(6,4), A7=(1,2), A8=(4,9). Initial centres are A1, A4 and A7.

Initial centroids: C1=(2,10), C2=(5,8), C3=(1,2). Assign each point to the nearest initial centroid using Euclidean distance. A point equidistant from two centres should be assigned using a stated tie-breaking rule.

Using nearest-centroid assignment, the clusters after assignment are:

• C1: A1

• C2: A3, A4, A5, A6, A8

• C3: A2, A7

Updated C1 centroid = (2/1, 10/1) = (2.00, 10.00).

Updated C2 centroid = (30/5, 30/5) = (6.00, 6.00).

Updated C3 centroid = (3/2, 7/2) = (1.50, 3.50).

3. Confusion matrix

A confusion matrix compares actual class labels with predicted class labels. For binary classification:

	Predicted positive	Predicted negative
Actual positive	TP	FN
Actual negative	FP	TN

TP: actual positive correctly predicted positive. TN: actual negative correctly predicted negative. FP: actual negative incorrectly predicted positive. FN: actual positive incorrectly predicted negative.

Evaluation formulas

```
Precision = TP/(TP+FP)
Recall = TPR = Sensitivity = TP/(TP+FN)
Specificity = TN/(TN+FP)
FPR = FP/(FP+TN) = 1 − Specificity
F1 = 2 × Precision × Recall / (Precision + Recall)
```

Precision answers: of predicted positives, what fraction are truly positive? Recall/sensitivity answers: of actual positives, what fraction were detected? Specificity answers: of actual negatives, what fraction were correctly identified? F1 is the harmonic mean of precision and recall.

4. Teacher's 2025 multiclass confusion matrix numerical

For one-versus-rest calculations, treat each class in turn as positive; all other classes are negative. Total observations = 818.

Class A: TP=150, FN=35, FP=22, TN=611; precision=0.8721; TPR/recall=0.8108; specificity=0.9652; FPR=0.0348; F1=0.8403.

Class B: TP=200, FN=37, FP=47, TN=534; precision=0.8097; TPR/recall=0.8439; specificity=0.9191; FPR=0.0809; F1=0.8264.

Class C: TP=350, FN=46, FP=49, TN=373; precision=0.8772; TPR/recall=0.8838; specificity=0.8839; FPR=0.1161; F1=0.8805.

5. ROC curve

The Receiver Operating Characteristic (ROC) curve plots True Positive Rate (TPR) on the vertical axis against False Positive Rate (FPR) on the horizontal axis at different classification thresholds.

```
TPR = TP/(TP+FN)
FPR = FP/(FP+TN)
```

At each threshold, predict positive when the score is at least the threshold, unless instructed otherwise. Count TP, FP, FN and TN, calculate TPR and FPR, and plot (FPR, TPR).

Interpretation

• Perfect classifier: (FPR=0, TPR=1), top-left corner. • Random/no-skill reference: diagonal from (0,0) to (1,1). • Always-negative classifier: (0,0). • Always-positive classifier: (1,1). A curve closer to the top-left generally indicates better discrimination.

Teacher's ROC dataset — thresholds and coordinates

There are 5 positive and 5 negative actual cases. At threshold 1, no score is ≥1, so TPR=0 and FPR=0. At threshold 0.9, only patient 1 is positive: TP=1, FP=0, so TPR=0.2 and FPR=0. Continue in the same way:

Threshold	TP	FP	FN	TN	TPR	FPR	(FPR,TPR)
1.0	0	0	5	5	0.00	0.00	(0.00,0.00)
0.9	1	0	4	5	0.20	0.00	(0.00,0.20)
0.7	2	1	3	4	0.40	0.20	(0.20,0.40)
0.5	3	3	2	2	0.60	0.60	(0.60,0.60)
0.3	4	3	1	2	0.80	0.60	(0.60,0.80)
0.1	5	5	0	0	1.00	1.00	(1.00,1.00)

Note: The threshold values 0.7, 0.5, 0.3 and 0.1 are applied as score ≥ threshold. The resulting points should be plotted in increasing FPR order; points can repeat FPR while TPR increases.

ROC inference for this dataset: the threshold changes the trade-off between detecting positives and false alarms. Lowering the threshold generally increases both TPR and FPR. This small dataset gives a coarse curve; avoid claiming perfect classification.

Practice Set — Questions to Rehearse

Module 3: Regression

- Define regression and explain its types.
- Explain OLS and derive the slope/intercept formulas.
- Solve the teacher's two regression questions without looking at the worked solutions.
- Define Pearson's correlation coefficient and differentiate correlation from regression.
- Explain multiple linear regression and write its matrix OLS estimate.

Module 4: Neural networks

- Define an artificial neuron and derive its output function for zero bias.
- Explain step, sigmoid, tanh, ReLU and softmax activation functions.
- Explain network topology and the functions of input, hidden and output layers.
- Explain backpropagation with a diagram and weight/bias update equations.
- Discuss the strengths and weaknesses of backpropagation.

Module 5: Clustering and evaluation

- Explain K-means and solve one epoch for the eight-point PYQ.
- Explain WCSS and the Elbow Method with a graph.
- Construct a confusion matrix and calculate precision, recall, sensitivity, specificity and F1.
- Solve the teacher's 3-class confusion matrix using one-versus-rest calculations.
- Plot the teacher's ROC curve for thresholds 1, 0.9, 0.7, 0.5, 0.3 and 0.1.
- Explain the ROC diagonal and the perfect, always-positive and always-negative classifiers.

Last-minute checklist

- I can write OLS objective and slope/intercept formulas from memory.
- I can solve both regression datasets and handle the regression direction correctly.
- I can draw and label an artificial neuron.
- I can explain backpropagation and write weight/bias update equations.
- I can calculate every confusion-matrix metric correctly.
- I can produce ROC coordinates from thresholds and plot them.
- I can perform K-means assignment and centroid updates.
- I have checked whether SVM is included in the confirmed exam portions.

End of notes — prioritise teacher-paper questions and practise calculations by hand. A recurring PYQ is useful evidence for revision priority, not a guarantee of repetition.